Week 1 Surprise Test

Cohort 2 | Week 1 | 50 minutes | 35 items in 6 parts

What this paper is for

This paper checks whether you can use Week 1's ideas cold, with no notes and no assistant: Python's lists, dictionaries and strings, cleaning data with pandas, NumPy, descriptive statistics and distributions, the central limit theorem and the z-test, and turning an analysis into a decision. Every item puts the business question first and the technique second, which is the order interviewers use, and several rebuild a well-known data interview question around an exhibit, so each one rewards the reasoning an interviewer listens for.

Rules

Rule	Detail
Time	50 minutes in one sitting for 35 items. The parts add up to 46.5 minutes, which leaves time for a last check of the answer sheet. Read this page and do step one before the clock starts. Each part gives its minutes as a guide; move on when an item is eating your time, and come back to it at the end.
Tools	Pen and this paper only: no laptop, no phone, no notes and no assistant. Every number can be worked by hand, and rough working goes in the margins.
Answers	Every answer goes on the answer sheet at the back, which is the page that is marked: one box for a lettered item, every correct box for a starred item (Q3, Q7, Q15 and Q33, each with at least one correct option), T or F for Q25 to Q27, a letter for each match item, and the number or the letters in order in the space. A starred item is right only when every correct box, and no other, is marked, and a number is right when it matches the key, rounded as the item asks. A wrong answer costs nothing, so answer every item.
Levels	Every item shows its level beside its number, easy, medium or hard. A hard item takes several steps of reasoning, never an obscure fact.
Exhibits	An exhibit is printed once, labelled by its part as Exhibit 2A, 2B, and every item that reads it follows it. Read code exactly as written, in Python 3 with pandas and NumPy.
Company	Kalpa Retail and its people are the course's fictional company. Swiggy, Zomato, Blinkit, Zepto, Uber and PhonePe appear in illustrative situations whose numbers are invented for the item, so no number here is a real result of these companies.
Afterwards	The paper is marked against the key, and the discussion opens on the items the room missed most.

Step one, before Part 1: rate yourself

On the answer sheet, rate yourself from 1 to 4 on each part before you read any item. Rate yourself as you are today: your rating beside your score in each part shows where confidence and skill disagree, and that gap is the first thing to practise.

1	2	3	4
I have not used this	I can follow it when someone shows me	I can do it alone on a small problem	I can find and fix a mistake in someone else's version

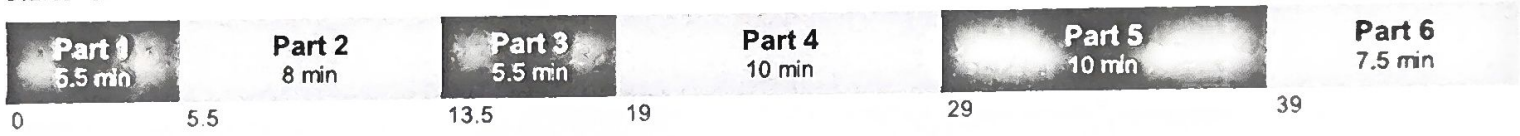
One rating for each of the six parts in the table below, in the boxes on the answer sheet.

The paper at a glance

Part	Area	Items	What it shows	Minutes	Easy	Medium	Hard
1	Python: lists, dictionaries and strings	Q1 to Q4 (4)	whether you can predict what lists, dictionaries and strings do before the code runs	5.5	1	1	2
2	pandas: cleaning an export	Q5 to Q9 (5)	whether you can clean an export with pandas and know which steps move the revenue figure	8	0	2	3
3	NumPy: shapes, masks and types	Q10 to Q13 (4)	whether you can reason about NumPy shapes, masks and types	5.5	0	3	1
4	Descriptive statistics and distributions	Q14 to Q23 (10)	whether you can choose the right summary and the right distribution, and reason with them	10	3	4	3
5	Inference: the CLT and the z-test	Q24 to Q31 (8)	whether you can use the central limit theorem and a z-test, and say what a result means	10	0	6	2
6	Analysis: from numbers to a decision	Q32 to Q35 (4)	whether you can turn an analysis into the decision it supports, with the right caveat	7.5	0	3	1
	Total	**35**		**46.5**	**4**	**19**	**12**

Pacing

Each band is as wide as its part's minutes; the timed items fill 46.5 of the 50 minutes, and the minute each part starts is under it.



Part 1. Does your Python do what you think it does with Kalpa's order records?

Four items, Q1 to Q4, about 5.5 minutes. What it shows: whether you can predict what lists, dictionaries and strings do before the code runs.

Kalpa Retail's app sends each order to the data team as a Python dictionary, and a day's orders arrive as a list of them. Exhibit 1A shows four orders from the Pune and Mumbai stores, exactly as they arrive, and

Q1 and Q2 each start from this list as printed. Kavya Nair, the senior analyst on your team, reviews every snippet before it reaches a report.

Exhibit 1A *Four orders as they arrive from Kalpa's app, in Python*

```
orders = [
    {"id": "KR-0101", "city": " Pune",  "amount": 1200},
    {"id": "KR-0102", "city": "pune",   "amount": 450},
    {"id": "KR-0103", "city": "Mumbai", "amount": 3100},
    {"id": "KR-0104", "city": "PUNE ",  "amount": 800},
]
```

Exhibit 1B *A teammate's what-if, run straight after Exhibit 1A.*

```
snapshot = orders.copy()
snapshot[0]["amount"] = 0
snapshot.append({"id": "KR-0105", "city": "Delhi", "amount": 500})
print(len(orders), orders[0]["amount"])
```

Q1 **Hard** · Predict the output · *one letter*

A teammate wants to try a what-if on a copy of the list, so the real orders stay untouched. What does the last line of Exhibit 1B print?

A. `4 1200`

B. `5 0`

C. `5 1200`

D. `4 0`

Exhibit 1C *Revenue by city, run in a fresh session on Exhibit 1A's list exactly as printed.*

```
totals = {}
for o in orders:
    city = o["city"].strip().title()
    totals[city] = totals.get(city, 0) + o["amount"]
print(totals)
```

Q2 **Medium** · Predict the output · *one letter*

Kavya asks for revenue by city. What does Exhibit 1C print?

A. `{'Pune': 2450, 'Mumbai': 3100}`

B. `{'Pune': 1650, 'Mumbai': 3100, 'PUNE': 800}`

C. `{'Pune': 3, 'Mumbai': 1}`

D. `{'Pune': 800, 'Mumbai': 3100}`

Q3* **Hard** · Choose every correct option · *every correct letter*

Kalpa's new order system writes every order id in the shape 'KR-YYYY-MM-NNNNNN', for example oid = 'KR-2026-04-000123', where the last six digits are the order's serial number with leading zeros. Which expressions return the serial as an integer, 123 here, for every id of this shape? Choose every correct option.

A. `int(oid.split("-")[2])`

B. `int(oid[-6:])`

C. `int(oid[-6:-1])`

D. `int(oid.split("-")[-1][-3:])`

E. `int(oid.split("-")[-1])`

Q4 **Easy** · True or false, with the reason · *one letter*

Statement: Kalpa can key a dictionary by a customer's list of order ids, for example gifts[['KR-0101', 'KR-0102']] = 'mug'. Choose the verdict and the reason.

A. True, because a dictionary accepts any Python object as a key.

B. True, because the list holds only strings, and strings can be keys.

C. False, because Python only accepts strings and integers as dictionary keys.

D. False, because a list can change after it is made, so it cannot be hashed.

Part 2. Can Kalpa trust the monsoon-sale export after pandas has cleaned it?

Five items, Q5 to Q9, about 8 minutes. What it shows: whether you can clean an export with pandas and know which steps move the revenue figure.

The first eight rows of Kalpa's monsoon-sale export reach your team as a CSV file, shown as Exhibit 2A. Amounts are rupees as the store system typed them, a negative quantity is a return, and order_id is the identity of an order: one order, one row. Anand Iyer, the finance controller, checks your revenue figure against the bank's settlements, so any error your cleaning hides will surface in his reconciliation.

Exhibit 2A *The first eight rows of monsoon_export.csv, read with pd.read_csv('monsoon_export.csv') and no other arguments. KR-503's city is empty in the file.*

order_id	customer_id	city	amount	qty	order_date
KR-501	C-2101	Pune	1,250	2	2026-07-03
KR-502	C-2102	pune	640	1	2026-07-03
KR-503	C-2103		980	1	2026-07-03
KR-504	C-2104	Mumbai	2,400	3	04/07/2026
KR-505	C-2105	Delhi	n/a	1	2026-07-04
KR-506	C-2106	Delhi	85,000	40	2026-07-05
KR-507	C-2107	Mumbai	-640	-1	2026-07-05

Q5 **Hard** · Choose the line · *one letter*

Amounts with a comma arrive as text. Which line turns df['amount'] into numbers without losing any amount the file actually holds?

A. `df['amount'] = pd.to_numeric(df['amount'].str.strip(), errors='raise')`

B. `df['amount'] = pd.to_numeric(df['amount'].str.replace(',', ''), errors='coerce')`

C. `df['amount'] = pd.to_numeric(df['amount'].str.strip(), errors='coerce')`

D. `df['amount'] = pd.to_numeric(df['amount'].str.replace(',', '.'), errors='coerce')`

Exhibit 2B *Two checks Kavya runs on Exhibit 2A's eight rows after Q5's line.*

```
a = len(df.drop_duplicates())
b = len(df.drop_duplicates(subset='order_id'))
```

Q6 **Hard** · Predict the counts · *one letter*

What are a and b?

A. `a = 8, b = 8`

B. `a = 7, b = 8`

C. `a = 7, b = 7`

D. `a = 8, b = 7`

Q7* **Hard** · Choose every correct option · *every correct letter*

With amounts converted by Q5's line and duplicates dropped with df = df.drop_duplicates(subset='order_id'), Anand asks which further steps would change the revenue total, df['amount'].sum(). None of the five raises an error. Choose every step that changes the total.

A. `df['amount'] = df['amount'].fillna(0)`

B. `df = df.dropna(subset=['city'])`

C. `df = df[df['qty'] > 0]`

D. `df['city'] = df['city'].str.strip().str.title()`

E. `df = df[df['amount'] < 50000]`

Q8 **Medium** · Make the call · *one letter*

KR-506 is a Delhi caterer's order of 40 units for Rs 85,000, about 35 times the next largest order in the export. Meera Raghavan, Kalpa's CEO, wants the monsoon-sale revenue figure for Monday's review. How should that figure handle KR-506?

A. Delete KR-506, since it sits far beyond 1.5 IQR and would inflate every average Meera sees.

B. Report revenue without KR-506 as the headline, and give the total including it in a footnote.

C. Keep it in revenue, flag it as a bulk order, and report the total with and without it.

D. Cap it at the 95th percentile of order values, so it still counts but stops distorting the total.

Q9 **Medium** · Write the number · *a number*

The full monsoon-sale export has 12,480 rows. The cleaning run removes 214 duplicate rows by order_id, sends a further 37 rows, whose amount cannot be read, to the rejects log, and keeps all 96 returns as their own rows. If nothing else changes, how many clean rows should the run report?

Part 3. Can a dark store's data run as arrays instead of loops?

Four items, Q10 to Q13, about 5.5 minutes. What it shows: whether you can reason about NumPy shapes, masks and types.

An analyst at a quick-commerce company such as Zepto (an illustrative situation) holds the orders per hour for three dark stores over four evening hours as a NumPy array. Exhibit 3A builds it: rows are stores S1, S2 and S3, and columns are hours 1 to 4. Q10 and Q11 run on this array.

Exhibit 3A *Orders per hour: rows are stores S1 to S3, columns are hours 1 to 4.*

```
import numpy as np
orders = np.array([[120,  95, 140, 110],
                   [ 80,  60,  75,  90],
                   [200, 180, 210, 190]])
```

Q10 **Medium** · Write the number · *a number*

What does orders[orders > 100].mean() print for Exhibit 3A's array? Give it to two decimal places.

Q11 **Hard** · Choose the line · *one letter*

The analyst wants each store's hourly orders minus that store's own four-hour average, so a quiet store and a busy store can be compared. Which line gives a 3 by 4 array of exactly that?

A. `orders - orders.mean(axis=1)[:, None]`

B. `orders - orders.mean(axis=1)`

C. `orders - orders.mean(axis=0, keepdims=True)`

D. `orders - orders.mean(axis=1).reshape(1, -1)`

Q12 **Medium** · True or false, with the reason · *one letter*

Statement: np.array([250, 499, '1200']).mean() returns 649.67, the average of the three amounts. Choose the verdict and the reason.

A. True, because NumPy reads '1200' as a number when the other values are numbers.

B. True, because NumPy builds an array of mixed types and averages the numeric ones.

C. False, because one string turns all three values into text, which has no mean.

D. False, because NumPy drops the string and returns 374.5, the mean of 250 and 499.

Q13 **Medium** · Choose the reason · *one letter*

On 10 lakh deliveries, an analyst's Python for loop computes every delivery's distance in about 9 seconds, and the same formula on NumPy arrays takes about 0.05 seconds (illustrative timings). Which reason explains most of the gap?

A. NumPy spreads the loop across every CPU core, so eight cores do the work eight times faster.

B. NumPy keeps the numbers side by side in memory and runs the loop in compiled code.

C. NumPy is written in C, so importing it speeds up any Python loop that touches its arrays.

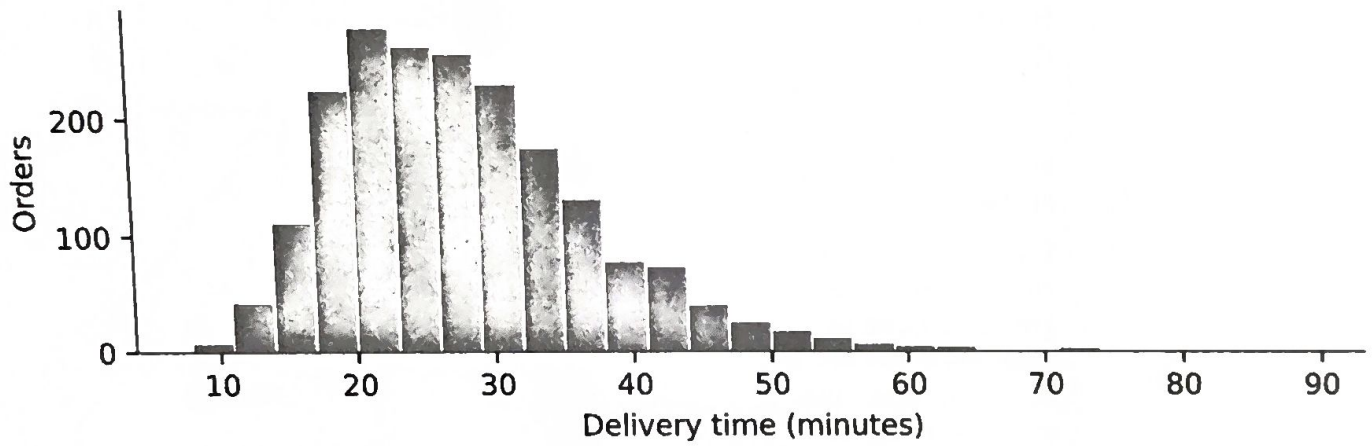
D. NumPy arrays are smaller than lists, so the data stay in memory instead of spilling to disk.

Part 4. What do the numbers say about a typical order, a typical wait and a typical tap?

Ten items, Q14 to Q23, about 10 minutes. What it shows: whether you can choose the right summary and the right distribution, and reason with them.

The situations below come from a food-delivery app such as Swiggy, a ride app such as Uber, a payments app such as PhonePe, a rice-packing line and one Kalpa store. Every number is invented for the item.

Exhibit 4A *Delivery times of 2,000 orders in one Bengaluru zone of a food-delivery app (illustrative).*



Q14 **Easy** · Read the shape · *one letter*

Which ordering of the mean, median and mode does the shape of Exhibit 4A imply?

A. mode < median < mean

B. mean < median < mode

C. mean = median = mode

D. median < mode < mean

Q15* Hard · Choose every correct option — *every correct letter*

Nine orders from one Kalpa store on a Tuesday, in rupees and sorted: 380, 390, 400, 420, 430, 450, 460, 510 and 47,000. The Rs 47,000 order is a school's bulk purchase. The trimmed mean here drops the single highest and the single lowest value, then averages the rest. Which of these statistics change by less than 10 percent when the bulk order is left out? Choose every correct option.

A. the mean order value

B. the median order value

C. the standard deviation

D. the trimmed mean

E. the range of order values

Match table 4 *For Q16 to Q20, write the letter of the most specific distribution that describes each quantity exactly; an approximation does not count. A letter can be used once, more than once or not at all.*

Item		Match		
Q16 Medium	How long a delivery partner waits at a restaurant: never below zero, mostly short, with a long right tail.		A	Bernoulli
Q17 Easy	How many of 20 PhonePe users shown a cashback banner tap it, when each taps independently with the same chance.		B	Binomial
Q18 Medium	Each rice packet's weight minus 1,002 grams, divided by the machine's true standard deviation, when the machine's packet weights follow a normal curve with mean 1,002 grams.		C	Normal
Q19 Easy	Whether the next PhonePe user shown the banner taps it, recorded as 1 or 0.		D	Standard normal
Q20 Hard	How many of 500 packets from that machine weigh under 1,000 grams, when each packet is underweight independently with the same chance.		E	None of these four

Write each letter on the answer sheet.

Q21 **Medium** · Write the number — *a number*

The banner of Q17 and Q19 converts each user independently with probability 0.2. Of five users who see it, what is the probability that exactly two tap it? Give it to three decimal places.

Q22 **Hard** · Choose the statement *one letter*

The payments team (illustrative) shows the banner to 400 users, with the same 0.2 chance each, and 104 tap it. Choose the statement that holds.

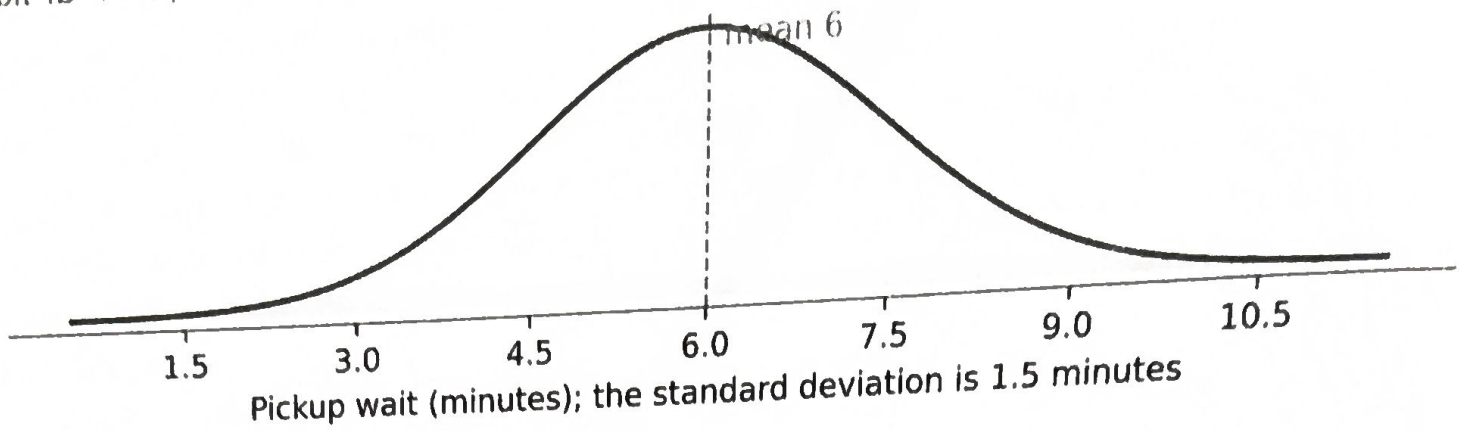
A. Expect 80 taps with a standard deviation of 64, so 104 lies well inside ordinary variation for 0.2.

B. Expect 80 taps with a standard deviation of 8, so 104 sits 3 standard deviations above: unusual.

C. Expect 80 taps with a standard deviation of 16, so 104 sits 1.5 standard deviations above: unremarkable.

D. Expect 100 taps with a standard deviation of 10, so 104 is about what a 0.2 chance would give.

Exhibit 4B *Pickup waits in one Hyderabad zone of a ride app (illustrative): a normal curve with mean 6 minutes.*



Q23 **Medium** · Read the curve · *one letter*

Using Exhibit 4B, about what share of riders wait more than 9 minutes for pickup?

A. About 16 percent

B. About 5 percent

C. About 2.5 percent

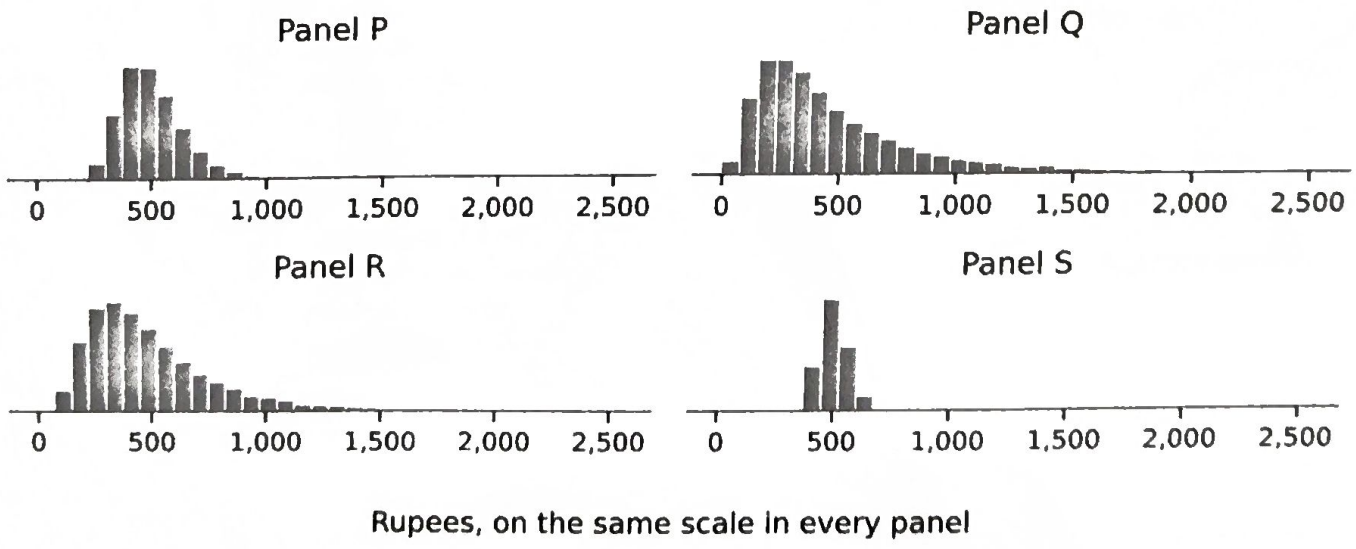
D. About 0.15 percent

Part 5. Is the gap real, or the noise of a sample?

Eight items, Q24 to Q31, about 10 minutes. What it shows: whether you can use the central limit theorem and a z-test, and say what a result means.

A food-delivery app's order values (illustrative) are skewed: most orders are small and a few are large. Exhibit 5A shows that population and, in a shuffled order, the averages of many random samples of 2, 10 and 50 orders drawn from it.

Exhibit 5A *One population of order values and the averages of samples of 2, 10 and 50 orders, in a shuffled order (illustrative).*



Q24 **Medium** · Read the panels · *one letter*

Which panel of Exhibit 5A shows the averages of samples of 10 orders?

A. Panel P

B. Panel Q

C. Panel R

D. Panel S

Q25 **Medium** · True or false · *T or F*

Statement: The order values in a random sample of 5,000 orders follow a normal curve, because the sample is large.

Q26 **Medium** · True or false · *T or F*

Statement: Even though single orders are skewed, the average of 100 random orders is close enough to normal for a z-test on the mean.

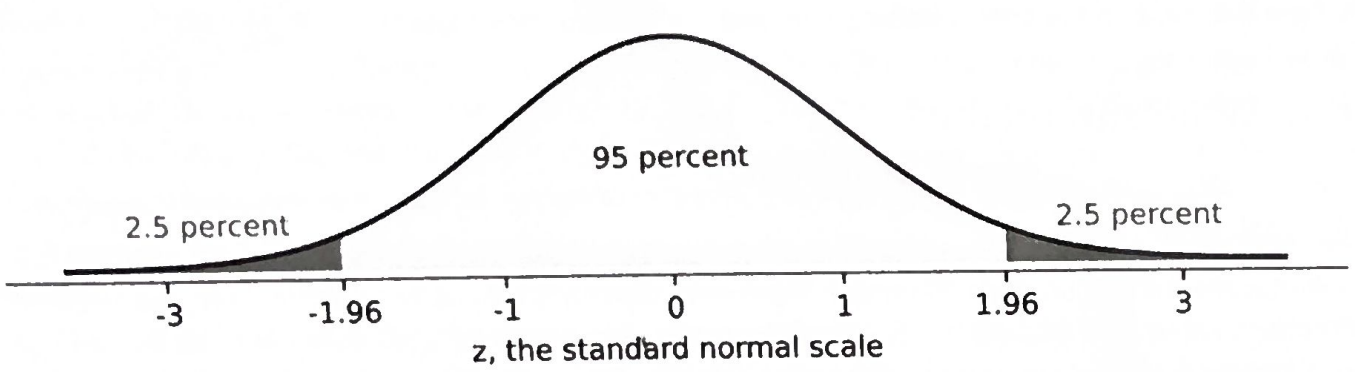
Q27 **Hard** · True or false · *T or F*

Statement: Doubling the number of orders sampled halves the standard error of the average.

Q28 **Medium** · Write the number · *a number*

A food-delivery app's zone manager states (illustrative) that the zone's mean delivery time is 30 minutes, and past data give a standard deviation of 8 minutes. A random sample of 100 deliveries averages 32.0 minutes. Compute the z statistic for this sample mean, to one decimal place.

Exhibit 5B *The standard normal curve, with the 5 percent beyond plus or minus 1.96 shaded, for a two-sided test at the 5 percent level.*



Q29 **Medium** · Make the call · *one letter*

A second zone of the same app makes the same 30-minute claim, with the same standard deviation of 8 minutes. A random sample of 64 of its deliveries averages 32.2 minutes, which gives z = 2.2. Using Exhibit 5B, what should this zone's manager be told at the 5 percent level, two-sided?

A. Keep the claim: the 2.2-minute gap is well inside one standard deviation of 8 minutes, so it is noise.

B. Reject the claim: z = 2.2 is beyond 1.96, so a sample mean this far from 30 is rare if it holds.

C. Reject the claim: z = 2.2 means there is only a 2.2 percent chance that the claim is true.

D. Keep the claim: z = 2.2 is far above 0.05, so the result does not reach significance.

Q30 Hard · Assertion and reason · *one letter*

Later the app tests the first zone's 30-minute claim on a random sample of 6.4 lakh deliveries from that zone's full history. The standard deviation is still 8 minutes, and the sample averages 30.05 minutes.

Assertion: The 0.05-minute gap, 3 seconds, is significant at the 5 percent level, two-sided.

Reason: A gap that is significant at the 5 percent level is large enough to matter to customers.

Choose one.

A. Both are true, and the reason explains the assertion.

B. Both are true, and the reason does not explain the assertion.

C. The assertion is true, and the reason is false.

D. The assertion is false, and the reason is true.

E. Both are false.

Q31 **Medium** · Name the error · *one letter*

A quick-commerce team (illustrative) tests, at the 5 percent level, whether new packaging changes the share of orders that arrive damaged. The test rejects 'no change', and the packaging goes to every store. If the packaging in truth changes nothing, which error was made, and what does it cost?

A. A Type I error, and the cost is a rollout that buys nothing.

B. A Type II error, and the cost is a rollout that buys nothing.

C. A Type I error, and the cost is better packaging the team never adopts.

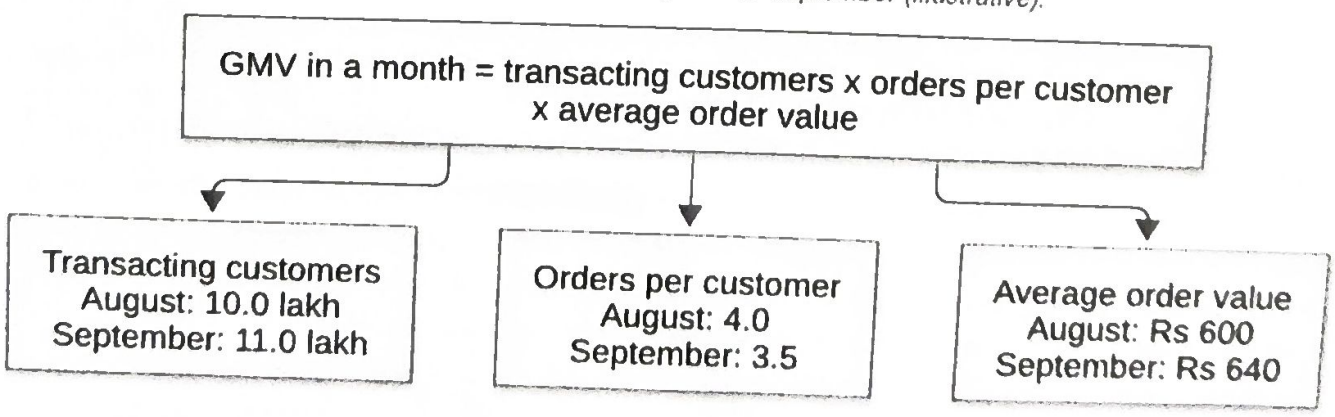
D. A Type II error, and the cost is better packaging the team never adopts.

Part 6. What should the business hear?

Four items, Q32 to Q35, about 7.5 minutes. What it shows: whether you can turn an analysis into the decision it supports, with the right caveat.

A quick-commerce app such as Blinkit (an illustrative situation) breaks its monthly GMV, the value of every order placed, into three branches: transacting customers, orders per customer and average order value. Exhibit 6A gives August and September.

Exhibit 6A *The GMV tree, with each branch's value in August and September (illustrative).*



Q32 **Medium** · Write the number · *a number*

By what percent did GMV change from August to September? Give one decimal place, with its sign.

Q33* **Hard** · Choose every correct option · *every correct letter*

The CEO reads the GMV change as healthy growth. Which of these readings does Exhibit 6A support? Choose every correct option.

A. Orders per customer fell 12.5 percent from August to September.

B. Orders per customer fell 14.3 percent from August to September.

C. The total number of orders fell, even though GMV rose.

D. At August's average order value, September's GMV would have fallen.

E. The average customer spent more in September than in August.

Q34 **Medium** · Put the steps in order · *letters in order*

An interviewer at a food-delivery company says: 'Repeat orders in Pune fell 8 percent last month. Walk me through your approach.' Put your first five moves in order.

A. Split the fall by customer segment, area and week, to see where it sits.

B. Confirm the metric: how a repeat order is counted, over which window and for whom.

C. Test the strongest hypothesis against data, and write the claim, evidence, caveat and action.

D. Check the data behind the number: missing days, duplicate orders, a logging change in the app.

E. List hypotheses for the part that holds the fall, each with the evidence that would settle it.

Write the letters in order on the answer sheet.

Q35 **Medium** · Choose the response · *one letter*

At a food-delivery app such as Zomato (illustrative), restaurants with more photos on their page get more orders: across 20,000 restaurants, the correlation is 0.6. The product team wants to push every restaurant to add photos. Choose the best response.

A. Push it now: a correlation of 0.6 across 20,000 restaurants is far too strong to be chance.

B. Test it first: ask the 1,000 restaurants with the fewest photos to add some, and compare their orders with the rest.

C. Drop the idea: a correlation between two numbers can never say anything useful about what causes what.

D. Test it first: pick at random which restaurants are asked to add photos, and compare with those not asked.